

Work, Energy and Power

Complete chapter notes — concepts, every derivation worked in full, standard results, worked examples and the traps that cost marks.

The spine of this chapter: work is the transfer of energy, and everything else is bookkeeping. $W_{\text{net}} = \Delta K$ holds *always*, with no exceptions. Add potential energy to the ledger and you get conservation of mechanical energy — but only when the forces doing work are conservative.

1 · Work

1.1 Definition

$W = \vec{F} \cdot \vec{d} = Fd \cos \theta$ — a **scalar**. Unit: joule (J) = N m. θ is the angle between the force and the **displacement**.

Work is done only when a force produces a displacement *and* has a component along it. Holding a heavy bag stationary does no work in the physics sense, however tired you get.

1.2 Sign of work

Angle	Sign	Standard examples
$\theta < 90^\circ$	Positive	Applied force on a moving block; gravity on a falling body
$\theta = 90^\circ$	Zero	Centripetal force ; normal reaction on a horizontal floor; tension on a bob in circular motion; the force on a body carried horizontally
$\theta > 90^\circ$	Negative	Friction on a sliding block; gravity on a body moving upward

FREQUENTLY ASKED

"Why is no work done by the centripetal force?" Because it acts perpendicular to the instantaneous displacement at every point, so $W = Fs \cos 90^\circ = 0$. The force changes the *direction* of the velocity, never its magnitude — and it is only the magnitude that carries kinetic energy.

1.3 Work done by a variable force

When F changes with position, split the displacement into pieces so small that F is effectively constant across each one. Over a piece dx the work is $dW = F dx$, and the total is

the sum:

$W = \int_{x_1}^{x_2} F dx$ = the **area under the F - x graph** between x_1 and x_2 . Area below the axis counts as negative work.

2 · Kinetic energy

$K = \frac{1}{2}mv^2 = \frac{p^2}{2m}$ and $p = \sqrt{2mK}$ — a scalar, always positive, measured in joules.

2.1 A pattern worth spotting

Three results in your syllabus share one algebraic shape:

Result	Where it comes from
$W = \frac{F^2}{2k}$	Work done stretching a spring by a <i>given force</i> F
$K = \frac{p^2}{2m}$	Linear kinetic energy in terms of momentum
$K_{\text{rot}} = \frac{L^2}{2I}$	Rotational kinetic energy in terms of angular momentum (Ch 6)

All three have the form **(quantity)² ÷ (2 × inertia-like term)**. In every case, if the numerator is held fixed, the answer is **inversely proportional** to the denominator.

EXAM APPLICATION

Two springs stretched by the *same force*, with $k_A > k_B$: since $W \propto 1/k$, we get $W_A < W_B$ — **more work is done on the softer spring B**. Two bodies with *equal angular momenta*, $I_A > I_B$: since $K \propto 1/I$, we get $K_A < K_B$ — **body B has the greater kinetic energy**. These are the same question in two costumes.

Also worth carrying: for **equal momenta**, the lighter body has more KE; for **equal KE**, the heavier body has more momentum.

3 · The Work-Energy Theorem

Statement. The work done by the **net force** acting on a body equals the change in its kinetic energy. $W_{\text{net}} = \Delta K = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$

DERIVATION 1 Work-energy theorem — constant force, then variable force

Part A — constant force (1-D).

1. Start from the third equation of motion: $v^2 = u^2 + 2as$.
2. Multiply throughout by $\frac{1}{2}m$: $\frac{1}{2}mv^2 - \frac{1}{2}mu^2 = mas$.
3. But $ma = F$ and $Fs = W$, so the right-hand side is exactly the work done.
4. $\Rightarrow W = \frac{1}{2}mv^2 - \frac{1}{2}mu^2 = \Delta K$.

Part B — variable force. This is the version usually asked, because it needs calculus.

1. For a small displacement dx the work done is $dW = F dx$, so the total work is $W = \int_{x_1}^{x_2} F dx$.
2. By Newton's second law $F = ma = m \frac{dv}{dt}$.
3. Use the chain rule to change the variable: $\frac{dv}{dt} = \frac{dv}{dx} \cdot \frac{dx}{dt} = v \frac{dv}{dx}$.
4. Substitute: $dW = \left(mv \frac{dv}{dx} \right) dx = mv dv$.
5. Integrate between the initial and final speeds: $W = \int_u^v mv dv = m \left[\frac{v^2}{2} \right]_u^v$.
6. $\Rightarrow W = \frac{1}{2}mv^2 - \frac{1}{2}mu^2 = K_f - K_i$.

$W_{\text{net}} = \Delta K$. **Positive net work $\Rightarrow$ the body speeds up; negative $\Rightarrow$ it slows down; zero $\Rightarrow$ the speed is unchanged.**

WHERE THIS LOSES MARKS

1. The theorem uses the **net** work of *all* forces — gravity and friction included — not just the applied force.
2. If the question says "for a variable force", you must show the $\int F dx = \int mv dv$ route. The $v^2 = u^2 + 2as$ argument alone will not earn full marks.

4 · Potential energy

Potential energy is energy stored by virtue of position or configuration. It is defined only for **conservative** forces.

4.1 Gravitational PE near the Earth's surface

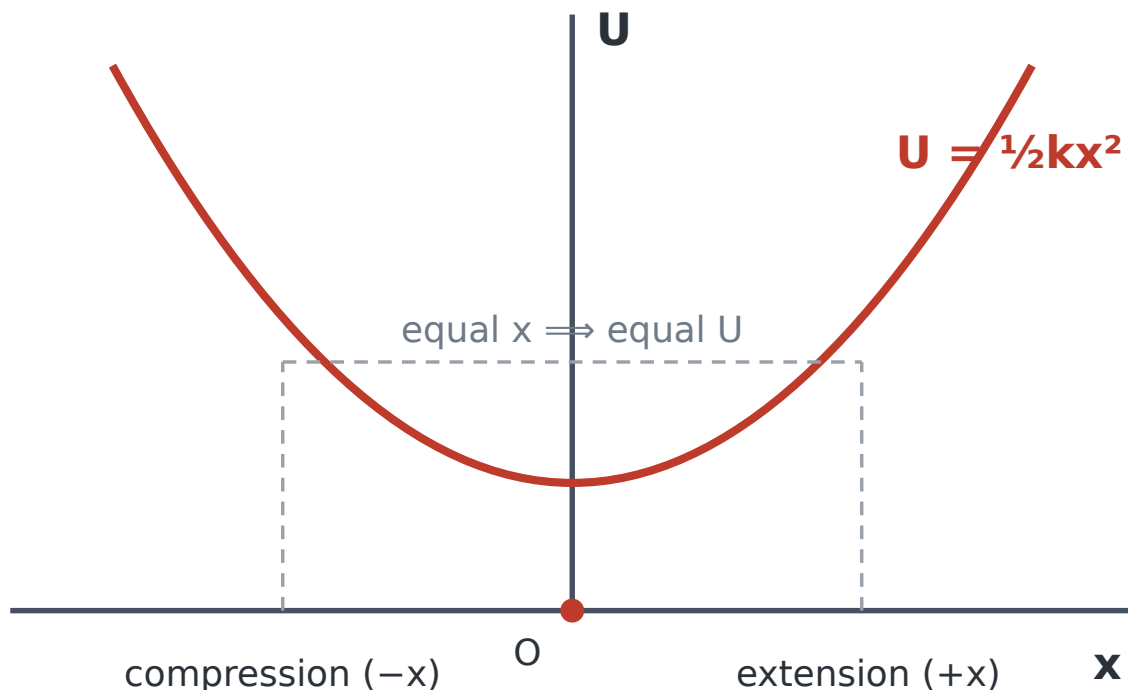
$U = mgh$, taking g as constant over the height involved. Only *changes* in U are physically meaningful, so you may set the zero level wherever it is convenient.

4.2 Elastic potential energy of a spring

DERIVATION 2 Potential energy of a spring, $U = \frac{1}{2}kx^2$

1. **Hooke's law.** When a spring is stretched or compressed by x from its natural length, it exerts a **restoring** force $F_s = -kx$. The minus sign says the force opposes the displacement; k is the spring constant, in N m^{-1} .
2. To stretch the spring *slowly* (no kinetic energy gained), an external agent must apply an equal and opposite force $F_{\text{ext}} = +kx$ at every instant.
3. Work done by the agent over a further small stretch dx : $dW = kx dx$.
4. Total work from the natural length to extension x : $W = \int_0^x kx dx = k \left[\frac{x^2}{2} \right]_0^x = \frac{1}{2}kx^2$.
5. This work is stored in the spring as elastic potential energy: $U(x) = \frac{1}{2}kx^2$. (The spring itself does $-\frac{1}{2}kx^2$ of work.)
6. Geometrically it is the area of the triangle under the F - x line: $\frac{1}{2} \times x \times kx$.

$U(x) = \frac{1}{2}kx^2$. **Work done in going from x_1 to x_2 is $\frac{1}{2}k(x_2^2 - x_1^2)$ — not $\frac{1}{2}k(x_2 - x_1)^2$.**



Spring PE against displacement. An **upward parabola**, symmetric about the mean position, with its minimum ($U = 0$) at $x = 0$. The symmetry says that stretching and compressing by the same amount store the same energy. A stiffer spring (larger k) gives a narrower, steeper parabola.

4.3 Conservative and non-conservative forces

	Conservative	Non-conservative
Examples	Gravity, spring force, electrostatic force	Friction, air resistance, viscous drag
Work done	Path-independent — depends only on the endpoints	Path-dependent
Round a closed loop	Zero	Non-zero
Potential energy	Can be defined	Cannot be defined

For a conservative force in one dimension,

$F = -\frac{dU}{dx}$ — the force is the negative slope of the PE curve. Where U is a **minimum** the equilibrium is **stable**; where it is a maximum, unstable.

5 · Conservation of mechanical energy

$E = K + U = \text{constant}$, provided the forces doing work are **conservative**. With friction present: $\Delta K + \Delta U = W_{\text{non-conservative}}$.

DERIVATION 3 Conservation of mechanical energy for a freely falling body

A body of mass m is released from rest at height H . Take the ground as the reference level ($U = 0$) and ignore air resistance, so gravity is the only force doing work.

- At the top.** $v = 0$, so $K = 0$; $U = mgH$. Total $E = 0 + mgH = mgH$.
- At a depth x below the top** (height $H - x$). Using $v^2 = u^2 + 2gx$ with $u = 0$: $v^2 = 2gx$.
 $K = \frac{1}{2}m(2gx) = mgx$ and $U = mg(H - x)$.
 Total $E = mgx + mgH - mgx = mgH$.
- At the ground.** $v^2 = 2gH$, so $K = \frac{1}{2}m(2gH) = mgH$; $U = 0$. Total $E = mgH$.
- The total is the same at all three points, so mechanical energy is conserved.
 Potential energy is steadily converted into kinetic energy.

$E = mgH$ throughout. **Halfway down ($x = H/2$):** $K = U = mgH/2$. **The speed on reaching the ground is $\sqrt{2gH}$ — independent of the mass.**

6 • Power

$$P_{\text{avg}} = \frac{W}{t} \quad \cdot \quad P_{\text{inst}} = \frac{dW}{dt} = \vec{F} \cdot \vec{v} = Fv \cos \theta \quad \text{— unit watt (W) = J s}^{-1}.$$

- 1 horsepower = 746 W
- 1 kWh = 3.6×10^6 J — this is a unit of **energy**, not power. The "per hour" is part of the unit, not a rate.

7 • Collisions

Momentum is conserved in every collision. Kinetic energy is conserved **only** in an elastic collision. Write this sentence in any collision answer — it is often a mark on its own.

During a collision the internal forces are enormous and the contact time tiny, so external forces (gravity, friction) contribute negligible impulse. That is why momentum conservation holds even on a rough floor.

7.1 Elastic collision in one dimension

DERIVATION 4 Elastic collision in 1-D — final velocities

Masses m_1, m_2 move along one line with initial velocities u_1, u_2 and final velocities v_1, v_2 .

1. **Conservation of momentum:** $m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$

Group the like masses: $m_1(u_1 - v_1) = m_2(v_2 - u_2) \quad \dots \text{ (i)}$

2. **Conservation of kinetic energy:** $\frac{1}{2} m_1 u_1^2 + \frac{1}{2} m_2 u_2^2 = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2$

Cancel the $\frac{1}{2}$ and group: $m_1(u_1^2 - v_1^2) = m_2(v_2^2 - u_2^2)$

Factorise both differences of squares: $m_1(u_1 - v_1)(u_1 + v_1) = m_2(v_2 - u_2)(v_2 + u_2) \quad \dots \text{ (ii)}$

3. **Divide (ii) by (i)** — the masses and the bracketed differences cancel:

$$u_1 + v_1 = u_2 + v_2 \Rightarrow u_1 - u_2 = v_2 - v_1 \quad \dots \text{ (iii)}$$

In words: **relative velocity of approach = relative velocity of separation.** This is the signature of an elastic collision.

4. Solve (i) and (iii) as simultaneous equations in v_1 and v_2 .

$$v_1 = \left(\frac{m_1 - m_2}{m_1 + m_2} \right) u_1 + \left(\frac{2m_2}{m_1 + m_2} \right) u_2$$

$$v_2 = \left(\frac{m_2 - m_1}{m_1 + m_2} \right) u_2 + \left(\frac{2m_1}{m_1 + m_2} \right) u_1$$

The three special cases (with $u_2 = 0$)

Case	v_1	v_2	Physical picture
$m_1 = m_2$	0	u_1	The velocities are exchanged — carrom, Newton's cradle. The first body stops dead, the second moves off with the first one's speed.
$m_1 \gg m_2$ heavy hits light	$\approx u_1$	$\approx 2u_1$	The heavy body ploughs on almost unaffected; the light body shoots off at twice the speed.
$m_1 \ll m_2$ light hits heavy	$\approx -u_1$	≈ 0	The light body rebounds with almost the same speed ; the massive body stays essentially at rest. (A ball bouncing off a wall.)

FREQUENTLY ASKED

"What happens when a light body collides elastically with a massive stationary body?"

Substitute $m_1 \ll m_2$ and $u_2 = 0$ into the results: $v_1 \approx -u_1$ and $v_2 \approx 0$. The light body **rebounds with its original speed reversed**, and the massive body remains at rest. Its kinetic energy is essentially unchanged.

7.2 Perfectly inelastic collision

DERIVATION 5 Perfectly inelastic collision — common velocity and energy lost

1. m_1 moving with u_1 strikes m_2 at rest, and the two stick together and move with a common velocity v .

2. Momentum: $m_1 u_1 = (m_1 + m_2) v \Rightarrow v = \frac{m_1 u_1}{m_1 + m_2}$.

3. Initial KE: $K_i = \frac{1}{2} m_1 u_1^2$. Final KE: $K_f = \frac{1}{2} (m_1 + m_2) v^2 = \frac{m_1^2 u_1^2}{2(m_1 + m_2)}$.

4. $\Delta K = K_i - K_f = \frac{1}{2} m_1 u_1^2 \left[1 - \frac{m_1}{m_1 + m_2} \right]$.

$$v = \frac{m_1 u_1}{m_1 + m_2} \quad \cdot \quad \Delta K_{\text{lost}} = \frac{1}{2} \frac{m_1 m_2}{m_1 + m_2} u_1^2 > 0 \quad \cdot \quad \text{fraction lost} = \frac{m_2}{m_1 + m_2}$$

The lost energy goes into heat, sound and permanent deformation. Momentum is still exactly conserved.

7.3 Coefficient of restitution

$e = \frac{\text{relative velocity of separation}}{\text{relative velocity of approach}} = \frac{|v_2 - v_1|}{|u_1 - u_2|}$ — dimensionless. $e = 1$ perfectly elastic, $e = 0$ perfectly inelastic, $0 < e < 1$ partially elastic.

A ball dropped from height h rebounds to $h' = e^2 h$; after n bounces, $h_n = e^{2n} h$. So $e = \sqrt{h'/h}$.

7.4 Elastic collision in two dimensions

With m_2 initially at rest and the bodies flying off at angles θ_1 and θ_2 :

- x -momentum: $m_1 u_1 = m_1 v_1 \cos \theta_1 + m_2 v_2 \cos \theta_2$
- y -momentum: $0 = m_1 v_1 \sin \theta_1 - m_2 v_2 \sin \theta_2$
- Energy: $\frac{1}{2} m_1 u_1^2 = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2$

Three equations, four unknowns — so one quantity must be supplied by the question.

DERIVATION 6 Equal masses in 2-D separate at 90°

1. Equal masses, target at rest. Momentum (as vectors): $\vec{u} = \vec{v}_1 + \vec{v}_2$.
2. Kinetic energy (the $\frac{1}{2}m$ cancels): $u^2 = v_1^2 + v_2^2$.
3. Square the vector equation: $u^2 = (\vec{v}_1 + \vec{v}_2) \cdot (\vec{v}_1 + \vec{v}_2) = v_1^2 + v_2^2 + 2\vec{v}_1 \cdot \vec{v}_2$.
4. Compare with step 2: the extra term must vanish, so $\vec{v}_1 \cdot \vec{v}_2 = 0$.

$\vec{v}_1 \perp \vec{v}_2$ — after an elastic collision between equal masses (one initially at rest), the two separate at 90° , provided both actually move.

8 • Motion in a vertical circle

A body of mass m is tied to a string of length R and whirled in a **vertical** circle. The speed is not constant — gravity does work as the height changes — so you need both a force equation and an energy equation.

DERIVATION 7 Vertical circle — minimum speeds and $T_{\text{bot}} - T_{\text{top}} = 6mg$

1. **At the highest point**, both the tension T_t and the weight mg point downward — toward the centre. So they together supply the centripetal force:

$$T_t + mg = \frac{mv_t^2}{R}$$

2. A string can only **pull**, never push, so the critical condition is $T_t = 0$ (the string just goes slack):

$$mg = \frac{mv_t^2}{R} \Rightarrow v_t^{\min} = \sqrt{gR}$$

3. **Energy conservation** from the top to the bottom — a drop of $2R$, with only gravity doing work:

$$\frac{1}{2}mv_b^2 = \frac{1}{2}mv_t^2 + mg(2R) \Rightarrow v_b^2 = v_t^2 + 4gR$$

4. Put $v_t^2 = gR$ for the critical case: $v_b^2 = gR + 4gR = 5gR \Rightarrow v_b^{\min} = \sqrt{5gR}$

5. **At the lowest point**, T_b points up (toward the centre) and mg points down:

$$T_b - mg = \frac{mv_b^2}{R}$$

6. Subtract the top equation from the bottom one:

$$T_b - T_t = \frac{m}{R} (v_b^2 - v_t^2) + 2mg = \frac{m}{R} (4gR) + 2mg$$

$$v_{\text{top}}^{\min} = \sqrt{gR} \quad \cdot \quad v_{\text{bottom}}^{\min} = \sqrt{5gR} \quad \cdot \quad v_{\text{side}}^{\min} = \sqrt{3gR}$$

$T_{\text{bot}} - T_{\text{top}} = 6mg$ — **and this holds at any speed, because $v_b^2 - v_t^2 = 4gR$ always follows from energy conservation.**

At the critical speed: $T_{\text{top}} = 0$, $T_{\text{bot}} = 6mg$, $T_{\text{side}} = 3mg$.

8.1 General position on the circle

Let u be the speed at the lowest point and θ the angle turned from there, so the height gained is $h = R(1 - \cos \theta)$:

$$v = \sqrt{u^2 - 2gR(1 - \cos \theta)} \quad \cdot \quad T = \frac{m}{R} [u^2 - 2gR(1 - \cos \theta)] + mg \cos \theta$$

Check the two ends: at $\theta = 0$ (bottom) this gives $T = \frac{mu^2}{R} + mg$; at $\theta = 180^\circ$ (top) it gives $T = \frac{mv_{\text{top}}^2}{R} - mg$. ✓

For a rod, or a bead inside a tube, the minimum speed at the top is **zero** — because a rod can push as well as pull, so it can support the body at the top with no speed at all.

FREQUENTLY ASKED

"A motorcyclist rides in a vertical circle. What is the necessary condition so that he does not fall?" His speed at the **highest point** must satisfy $v \geq \sqrt{rg}$, so that gravity is not more than what is needed to bend his path around the circle.

TRAPS THAT COST MARKS

1. In $W = Fd \cos \theta$, θ is **between the force and the displacement** — not between the force and the horizontal.
2. **The work-energy theorem uses NET work** of all forces. And if the question specifies a variable force, show the integral.
3. **Momentum is conserved in inelastic collisions too** — only kinetic energy is lost. Never try to conserve KE in an inelastic problem.
4. **Spring work from x_1 to x_2 is $\frac{1}{2}k(x_2^2 - x_1^2)$** , not $\frac{1}{2}k(x_2 - x_1)^2$.
5. **Minimum speed at the top of a vertical circle is $\sqrt{gR}$, not zero.** Zero applies only to a rod or a tube.
6. **1 kWh is energy, not power.**
7. **Heat produced by friction is $f \times (\text{relative sliding distance})$** , which is not the same as the work done by friction on one block when both bodies move.
8. **Equal masses in an elastic collision exchange velocities** — they do *not* both come to rest. That is a favourite false assertion in assertion-reason questions.

Chapter 5 of the rationalised NCERT (2023-24 onward). Every derivation above is written the way it should appear in the answer script: numbered steps, the key substitution shown explicitly, and the final result stated separately.